

Ordered Pairs in All Four Quadrants

Lesson 9-5

Name: _____

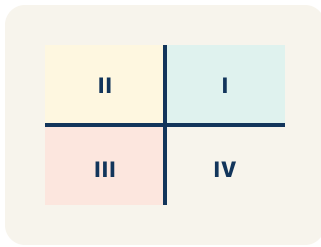
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



I (+, +) top-right, II (-, +) top-left, III (-, -) bottom-left, IV (+, -) bottom-right

Quadrant

Write the definition:

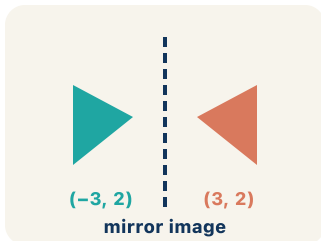
negative $\leftarrow$ 0



(-3, 2) means 3 left and 2 up from the origin

Negative coordinate

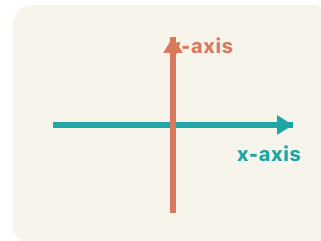
Write the definition:



(3, 2) reflected over the y-axis is (-3, 2) — same distance, opposite side

Reflection

Write the definition:

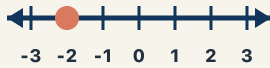


The x-axis goes left-right; the y-axis goes up-down; they cross at (0, 0)

Axis

Write the definition:

...-2, -1, 0, 1, 2...



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can plot ordered pairs in all four quadrants of the coordinate plane.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Quadrant

Negative coordinate

Reflection

Axis

Integer

1 One of the four regions of the coordinate plane, numbered I to IV, is a

_____.

2 A coordinate less than zero, which places a point left of or below the origin, is a

_____.

3 A flip of a point over an axis is a _____.

4 One of the two number lines, the x-axis or y-axis, that form the grid is an

_____.

5 A whole number or its opposite is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: In which quadrant is the point $(-4, 5)$?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

Step 1 The x-coordinate is negative (left) and the y-coordinate is positive (up).

Step 2 Left and up is Quadrant II.

 **Answer:** B. Quadrant II

 We do — together

Solve this one with your class using the same steps.

Problem: A point has a positive x-coordinate and a negative y-coordinate. Which quadrant is it in?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which point is located on the x-axis?

- A. $(4, 0)$
- B. $(0, 4)$
- C. $(4, 4)$
- D. $(-4, -4)$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A sonar ping locates an enemy submarine at $(-5, -2)$. In which quadrant should Captain Vega aim her shot?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

Show your work:

2. Captain Vega's lookout reports a ship in Quadrant II. What signs will its coordinates (x, y) have?

- A. $(+, +)$
- B. $(-, +)$
- C. $(-, -)$
- D. $(+, -)$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A point starts in Quadrant I at (4, 6). Describe a move that puts it in Quadrant III. What must happen to both coordinates? Can you give the exact new coordinates after your move?

Sentence starter: To move from Quadrant I to Quadrant III, both coordinates must become _____. Starting at (4, 6), I could move _____ units left and _____ units down to reach (____, ____), which is in Quadrant III because _____.

Show your work:

Reflect — Exit Ticket

A point has a negative x-coordinate and a negative y-coordinate. Which quadrant is it in?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Quadrant
2. **Notes 2:** Negative coordinate
3. **Notes 3:** Reflection
4. **Notes 4:** Axis
5. **Notes 5:** Integer
6. **We Try (notes):** D. Quadrant IV — *Positive x (right) and negative y (down) = Quadrant IV (bottom-right).*
7. **You Try (notes):** A. $(4, 0)$ — *A point is on the x -axis when its y -coordinate is 0. $(4, 0)$ is on the x -axis.*
8. **Try It 1:** C. Quadrant III — *A negative x means left and a negative y means down, so $(-5, -2)$ is in Quadrant III, where both coordinates are negative.*
9. **Try It 2:** B. $(-, +)$ — *Quadrant II is up and to the left of the origin, so x is negative and y is positive: $(-, +)$.*
10. **Exit Ticket:** C. Quadrant III — *Quadrant III is where both x and y are negative. This means the point is left of the origin and below the origin.*